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Department of Computer Science and Engineering

Academic Year 2025 – 2026 (Even Semester)

Degree, Semester & Branch: II Semester B.E. CSE

Course Code & Title: CS3452 – Theory of Computation

Name of the Faculty Member(s): Mrs. S. Manjula

Innovative Practice Description

Unit / Topic: Unit IV / Turing Machine – Design and Analysis

Course Outcome: CO 4

Topic Learning Outcome: TLO 15

Activity Chosen: Turing Machine Design Sprint

Date of Activity: 17.03.2026

Venue: CSE Classroom

Justification:

- The "Turing Machine Design Sprint" was designed to evaluate and strengthen students' analytical capacity to design and solve Turing Machine problems independently within a limited time frame.
- Turing Machine construction is a foundational topic in Theory of Computation that requires abstract reasoning about computation, tape operations, and state transitions — skills that are best tested through direct problem-solving activities.
- By assigning varied TM problems to students in a classroom setting, the instructor was able to gauge each student's depth of understanding and ability to apply Turing Machine concepts to new problems.
- The classroom discussion and random student presentation encouraged accountability, communication, and the ability to explain technical reasoning clearly to peers.
- Faculty elaboration at the end consolidated the learning and ensured all students left with accurate and complete understanding of the Turing Machine design principles.

Time Allotted for the Activity: 30 Minutes

Details of the Implementation:

The activity was conducted during a regular classroom session for II Semester B.E. CSE students as part of the innovative teaching initiative. The topic focused on the design and analysis of Turing Machines, a core concept in Unit IV of Theory of Computation. The session proceeded as follows:

Step 1: Problem Distribution

- Different Turing Machine problems were distributed to individual students across the classroom.
- Each student received a unique or distinct TM problem to solve, ensuring individual effort and broad coverage of problem types.

Step 2: Independent Problem Solving (Design Sprint)

- Students were given 30 minutes to independently analyze their assigned problem and construct the appropriate Turing Machine — defining states, tape symbols, transitions, initial and final states.
- This sprint-style format assessed each student's capacity to apply TM design knowledge under time constraints.

Step 3: Student Presentation

- After the sprint, a randomly selected student was called upon to present and explain their solution in front of the class on the board.
- This step promoted communication skills and peer learning, as classmates could observe and compare the presented solution with their own work.

Step 4: Faculty Elaboration and Consolidation

- The instructor reviewed and elaborated on the solution presented, highlighted common mistakes, and clarified the correct approach to Turing Machine design.
- This final step ensured conceptual accuracy and helped students who encountered difficulties to correct their understanding.

CO – PO / PSO Mapping:

CO	PO1	PO2	PO3	PO4	PO9	PO10	PSO1
CO 4	2	2	2	1	1	1	1

PO / PSO Mapped:

Innovative Practice	PO1	PO2	PO3	PO4	PO9	PO10	PSO1
Turing Machine Design Sprint	2	2	2	1	1	1	1

PO / PSO Justification:

PO / PSO	Justification
PO1	Students applied their knowledge of mathematics and engineering principles to analyse and construct Turing Machines, demonstrating their ability to translate abstract mathematical logic into a formal computational model.
PO2	Students identified the computational problem assigned to them, formulated the solution by representing the language through appropriate Turing Machine transitions, and arrived at substantiated conclusions by verifying the tape operations and acceptance conditions.
PO3	Students refined their understanding of Turing Machine architecture by designing detailed state transition diagrams and tape operation sequences, demonstrating their ability to move from high-level computational concepts to detailed design.
PO4	Students investigated the scope and complexity of their assigned Turing Machine problem by analysing the language structure, determining halting conditions, and evaluating whether the designed machine correctly accepted or rejected input strings.
PO9	Students gained problem-solving skills by independently designing Turing Machines within a sprint time frame, and further developed conflict resolution skills by comparing and discussing multiple approaches during the classroom presentation.
PO10	Students communicated their Turing Machine designs effectively by presenting their solutions in front of the class, explaining state transitions clearly, and comprehending the instructor's elaboration to refine their understanding.
PSO1	Students will identify the grammar and computational model for a language, design a Turing Machine to recognize it, and develop software solutions including mobile technologies with strong soft skills.

Images / Screenshot of the Practice:

Fig 1: Students solving Turing Machine problems independently during the Design Sprint

Fig 2: A randomly selected student presenting the TM solution on the board

Fig 3: Faculty elaborating and consolidating the Turing Machine design at the end of the session

Reflective Critique:

Feedback of Practice from Students and Other Stakeholders:

- Students appreciated the design sprint format as it challenged them to apply their knowledge independently and helped them identify gaps in their understanding of Turing Machine construction.
- The random presentation element encouraged all students to prepare thoroughly, as any student could be selected to present their solution to the class.
- Students found the faculty elaboration at the end particularly useful, as it clarified misconceptions and reinforced the correct approach to TM design.

Benefit of the Practice:

- The activity helped the instructor assess each student's individual understanding and problem-solving capacity for Turing Machine design in real time.
- Students developed the ability to construct Turing Machines independently and explain their reasoning clearly and confidently.

- The sprint format improved students' time management and analytical thinking skills under constrained conditions.

Challenges Faced in Implementation:

- A few students initially struggled with the time constraint and found it challenging to complete the TM construction within 30 minutes.
- These students were encouraged by the instructor to attempt partial solutions and were guided through the remaining steps during the faculty elaboration phase.

References:

- https://www.geeksforgeeks.org/turing-machine-in-toc/
- https://www.tutorialspoint.com/automata_theory/turing_machine.htm
- https://tilt.colostate.edu/TipsAndGuides/Tip/180
- https://www.edutopia.org/topic/collaborative-learning
- https://www.ritrjpm.ac.in/images/computer-science/28.CS8501_ColloborativeLearning.pdf

Signature of Faculty Member **HOD / CSE**